Lecture 3

- Conditional Independence
- Recursive Bayesian Estimation (static state)
- Markov Processes
- Recursive Bayesian Filter (dynamic state)

Recall Bayes' Rule:

given R.V.s

$$(X, Y) \sim p(x, y)$$

where

Y - measurement, observed

X - state, hidden

we want:

$$p(x|y) = \frac{p(y|x)\,p(x)}{\int_x p(y|x)\,p(x)\,dx}$$

$p(x|y)$: posterior; $p(y|x)$: measurement likelihood / sensor model; $p(x)$: prior; $\int_x p(y|x)\,p(x)\,dx$: normalization $= p(y)$

Q: What if we have a time sequence of measurements $Y_{1:t}$ and static state X? How to find posterior $p(x|y_{1:t})$ recursively given previous posterior $p(x|y_{1:t-1})$ and a new measurement y_t?

We have:

$$p(x|y_{1:t}) = \frac{p(y_{1:t}|x)\,p(x)}{\int_x \dots dx}$$

— trick to factor this dist.

Recall Independence:

Def: R.V.s Y_1, Y_2 are independent iff

$$p(y_1, y_2) = p(y_1)\,p(y_2)$$

Thm: The following are equivalent:

(i) Y_1, Y_2 are indep (aka $p(y_1, y_2) = p(y_1)\,p(y_2)$)

(ii) $p(y_1|y_2) = p(y_1)$

(iii) $p(y_2|y_1) = p(y_2)$

proof: (ii)

$$p(y_1, y_2) = p(y_1)\,p(y_2)$$

$$p(y_1|y_2)\,p(y_2) = p(y_1)\,p(y_2)$$

$$p(y_1|y_2) = p(y_1)$$

□

extend to arbitrary tuple:

Def: $Y_{1:t}$ are all mutually indep. iff $p(y_{1:t}) = \prod_{\tau=1}^{t} p(y_\tau)$

Conditional Independence

Def: Y_1 and Y_2 are conditionally independent given X iff

$$p(y_1, y_2|x) = p(y_1|x)\,p(y_2|x)$$

similarly, for a tuple, $Y_{1:t}$ are cond. indep. given X iff

$$p(y_{1:t}|x) = \prod_{\tau=1}^{t} p(y_\tau|x).$$

- Does not imply independence
- X is the common link between Y_1 and Y_2 (e.g. a common cause)
- If you know X, additionally knowing Y_2 gives no extra benefit to estimating Y_1

E.g. Puddles & Umbrellas

$Y_1 = \begin{cases} 0 & \text{no puddles} \\ 1 & \text{yes puddles} \end{cases}$ $Y_2 = \begin{cases} 0 & \text{no umbrellas} \\ 1 & \text{yes umbrellas} \end{cases}$

$X = \begin{cases} 0 & \text{not raining} \\ 1 & \text{yes raining} \end{cases}$

Y_1, Y_2 clearly not indep.

E.g. $p(y_1=1) = 0.01$

$p(y_1=1 \mid y_2=1) = 0.5$

but given X, Y_1, Y_2 are cond. indep.

E.g.

conditional indep.. $\begin{cases} p(y_1=1 \mid x=1) = 0.9 \\ p(y_1=1 \mid x=1, y_2=1) = 0.9 \end{cases}$

Thm: The following are equivalent

(i) Y_1, Y_2 are cond. indep. given X

(ii) $p(y_1 \mid x, y_2) = p(y_1 \mid x)$

(iii) $p(y_2 \mid x, y_1) = p(y_2 \mid x)$

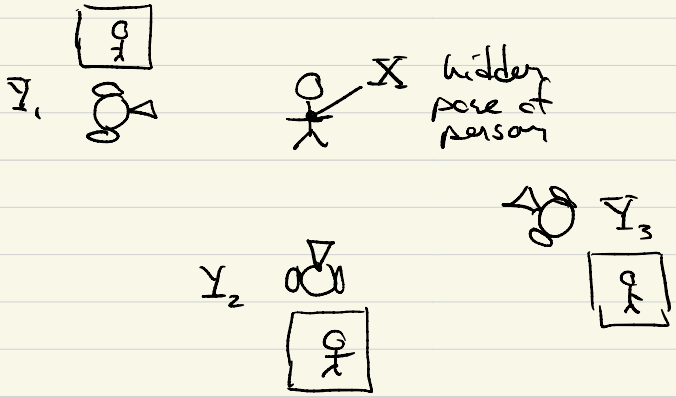
You will prove in PS1

Assumption (Conditionally Indep. Measurements):

Given a tuple of measurements $Y_{1:t}$, and a hidden state X,

$$p(y_{1:t} \mid x) = \prod_{\tau=1}^{t} p(y_\tau \mid x).$$

E.g. Robot with Camera



claim: cond. indep. measurements is a basic modeling assumption

$$p(y_1, y_2, y_3 \mid x) = \prod_{i=1}^{3} p(y_i \mid x).$$

Recursive Bayesian Estimation (Static state)

- consider measurement sequence $Y_{1:t}$, and fix hidden state X
- Assume cond. indep. measurements
- find an efficient recursive formula to find $p(x \mid y_{1:t})$ given $p(x \mid y_{1:t-1})$ and y_t

Bayes' Rule:

$$p(x \mid y_{1:t}) = \frac{p(y_{1:t} \mid x)\, p(x)}{\int_x p(y_{1:t} \mid x)\, p(x)\, dx}$$

by cond. indep. meas. we have

$$p(y_{1:t} \mid x) = p(y_t \mid x)\, p(y_{1:t-1} \mid x)$$

$$p(x \mid y_{1:t}) = \frac{p(y_t \mid x)\left(p(y_{1:t-1} \mid x)\, p(x)\right)}{\int_x p(y_t \mid x)\left(p(y_{1:t-1} \mid x)\, p(x)\right) dx}$$

divide by $\int_x p(y_{1:t-1} \mid x)\, p(x)\, dx$ on top and bottom

$$p(x \mid y_{1:t}) = p(y_t \mid x)\left[\frac{p(y_{1:t-1} \mid x)\, p(x)}{\int_x p(y_{1:t-1} \mid x)\, p(x) dx}\right] \Big/ \int_x p(y_t \mid x)\left[\frac{p(y_{1:t-1} \mid x)\, p(x)}{\int_x p(y_{1:t-1} \mid x)\, p(x) dx}\right] dx$$

by Bayes' Rule $= p(x \mid y_{1:t-1})$

posterior from previous iteration

we call this **prior**

we have

$$p(x \mid y_{1:t}) = \frac{p(y_t \mid x)\, p(x \mid y_{1:t-1})}{\int_x p(y_t \mid x)\, p(x \mid y_{1:t-1})\, dx}$$

Q: what if X is dynamic?

Random Processes (Stochastic Processes)

Def.: Collection of R.V.s indexed by time related by a joint distribution.

E.g. White Noise Process

all variables X_τ are mutually independent

$$(X_0, X_1, \dots, X_t) = (X_{0:t}) \sim p(x_{0:t}) = \prod_{\tau=0}^{t} p(x_\tau)$$

- we often consider "stationary" white noise: all $p(x_\tau)$ are the same prob. dist.
- we can also say X_τ are i.i.d.

E.g. Markov Process

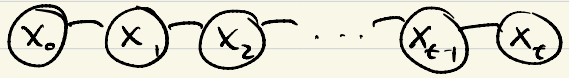
Def.: $(X_{0:t}) \sim p(x_{0:t})$ is Markov iff

(i) at t, $p(x_t | x_{0:t-1}) = p(x_t | x_{t-1})$

(ii) at $0 < \tau < t$, $p(x_\tau | x_{-\tau}) = p(x_\tau | x_{\tau-1}, x_{\tau+1})$ (↑ every time except ↑)

(iii) at 0, $p(x_0 | x_{1:t}) = p(x_0 | x_1)$

we think of a graphical representation (Graphical Model):



we'll see this graphical interpretation more next time!

Recursive Bayesian Filter (Dynamic Markov State)

predict:

prediction distribution $p(x_t | y_{1:t-1}) =$ "prior" (post. from $t-1$)

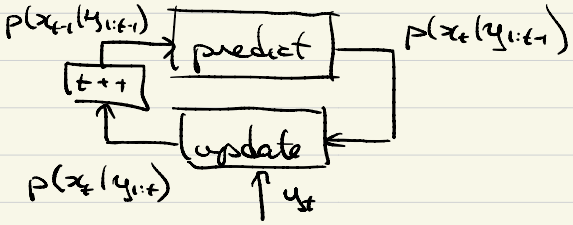
$$\int_{x_{t-1}} p(x_t | x_{t-1}) \, p(x_{t-1} | y_{1:t-1}) \, dx$$

$t \to t$

update:

posterior $p(x_t | y_{1:t}) = \dfrac{p(y_t | x_t) \, p(x_t | y_{1:t-1})}{\int_{x_t} p(y_t | x_t) \, p(x_t | y_{1:t-1}) \, dx_t}$

recursive structure:



Assumptions: (i) Cond. Indep. Meas.
(ii) Markovianity

proof: next time!

Discrete State:

suppose X_t, Y_t are discrete R.V.s

$x_t \in \{1, \dots, N\}$

$y_t \in \{1, \dots, M\}$

we can write $p(x_t)$ as a belief vector

$$\pi_t = \begin{bmatrix} p(x_t = 1) \\ p(x_t = 2) \\ \vdots \\ p(x_t = N) \end{bmatrix}$$

$$\mathbf{1}^T \pi_t = 1$$

def. Transition matrix

$$T = \begin{bmatrix} \vdots \\ \dots \; p(x_t = i \,|\, x_{t-1} = j) \; \dots \\ \vdots \end{bmatrix}$$

predict:

$$\boxed{\pi_t = T \pi_{t-1}}$$

update: PS1

when all operations Bayesian Filter become simple linear algebra expressions!